SMART INDIA HACKATHON 2025

TITLE PAGE

- **Problem Statement ID – 25076**
- **Problem Statement Title- AI-Based Farmer Query Support and Advisory System**
- **Theme- Agriculture, FoodTech & Rural Development**
- **PS Category- Software**
- **Team ID- 60538**
- **Team Name- CodeWiz Crew**

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❖ Proposed Solution

1.AI-Powered Conversational Support – Farmers ask queries in **Malayalam**, receive **real-time, context-aware answers** using **ChatGPT (GPT-5 mini)**, integrated with **expert-verified datasets** for **pest control, diseases, fertilizers, market trends, and schemes**.

2.Personalized Recommendations – Uses **location, crop type, soil data, and historical trends** to offer **tailored advice**, **crop/yield prediction**, and **weather forecasting** with **time-series ML models**.

3.Inclusive Access Channels – Available via **mobile apps (React Native), PWA, SMS, and email**, ensuring service in **low-connectivity areas**, overcoming **literacy barriers** with **voice assistance**.

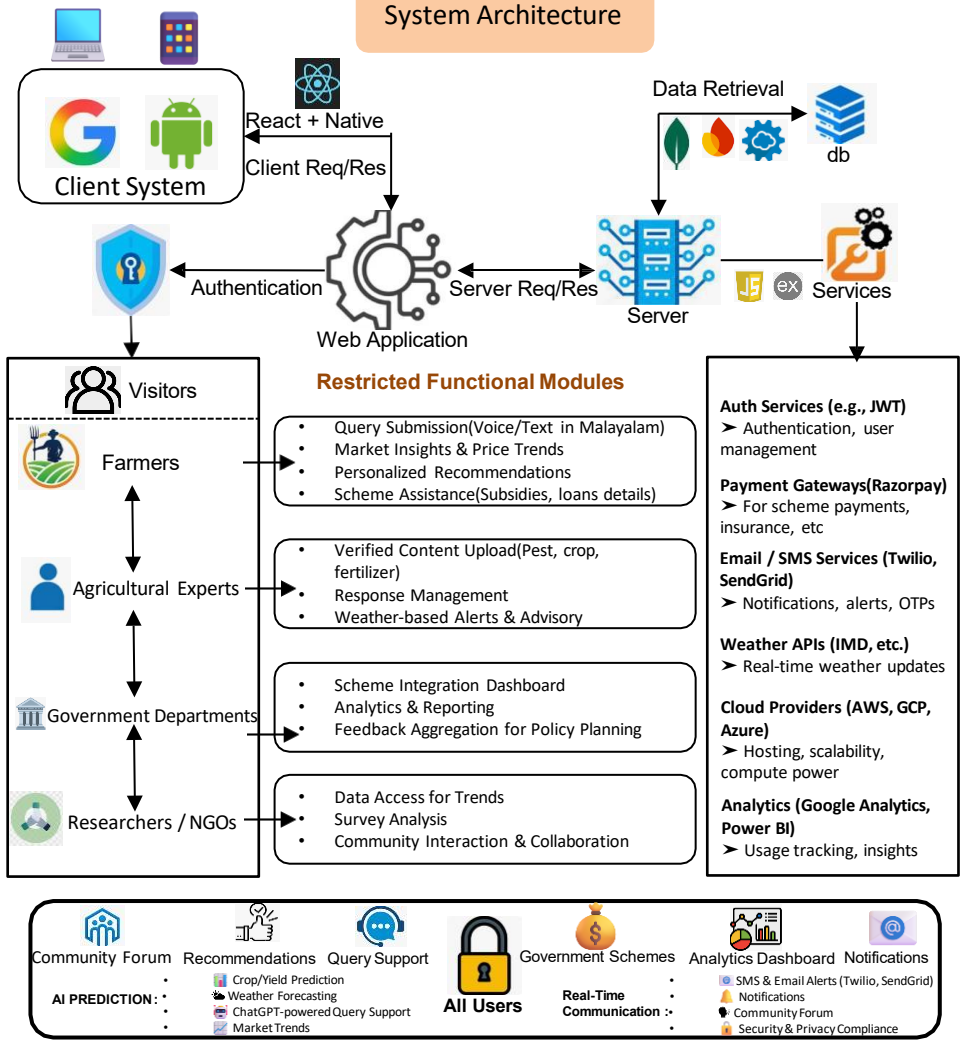
4.Government Scheme Integration – Provides details on **subsidies, insurance, and loan eligibility**, helping farmers access **financial aid** easily and on time.

5.Data-Driven Market Insights – Offers **price trends, demand forecasts, logistics guidance**, empowering farmers to make **informed selling decisions** and avoid exploitation.

6.Scalable, Secure Infrastructure – Built on **AWS/GCP/Azure**, with **Docker, Kubernetes, CI/CD, Terraform, SSL, GDPR compliance, RBAC**, and **audit trails** to ensure **data security and reliability**.

7.Community Feedback & Policy Support – Enables **peer interaction, upvotes, feedback loops**, and provides **aggregated analytics** for **policy planning**, helping governments and research bodies refine **agriculture strategies**.

System Architecture



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TECHNICAL APPROACH

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❖ Technology Stack

APIs :
- REST, Payment gateways,
- Geo, SMS/Email (Twilio,
- SendGrid), Weather APIs

Security
- SSL, GDPR, RBAC,
- Encryption, Backups
- Audit Trails, Blockchain Verification, MFA

Cloud & Deployment
- AWS, GCP, Azure,
- Docker, Kubernetes,
- Terraform, CI/CD,
- NGINX, Prometheus

Analytics
- Power BI, Tableau, Kafka
- Spark, Google Analytics
- BigQuery, Metabase,
- SSR & Feedback Reporting

AI-powered Farmer Support

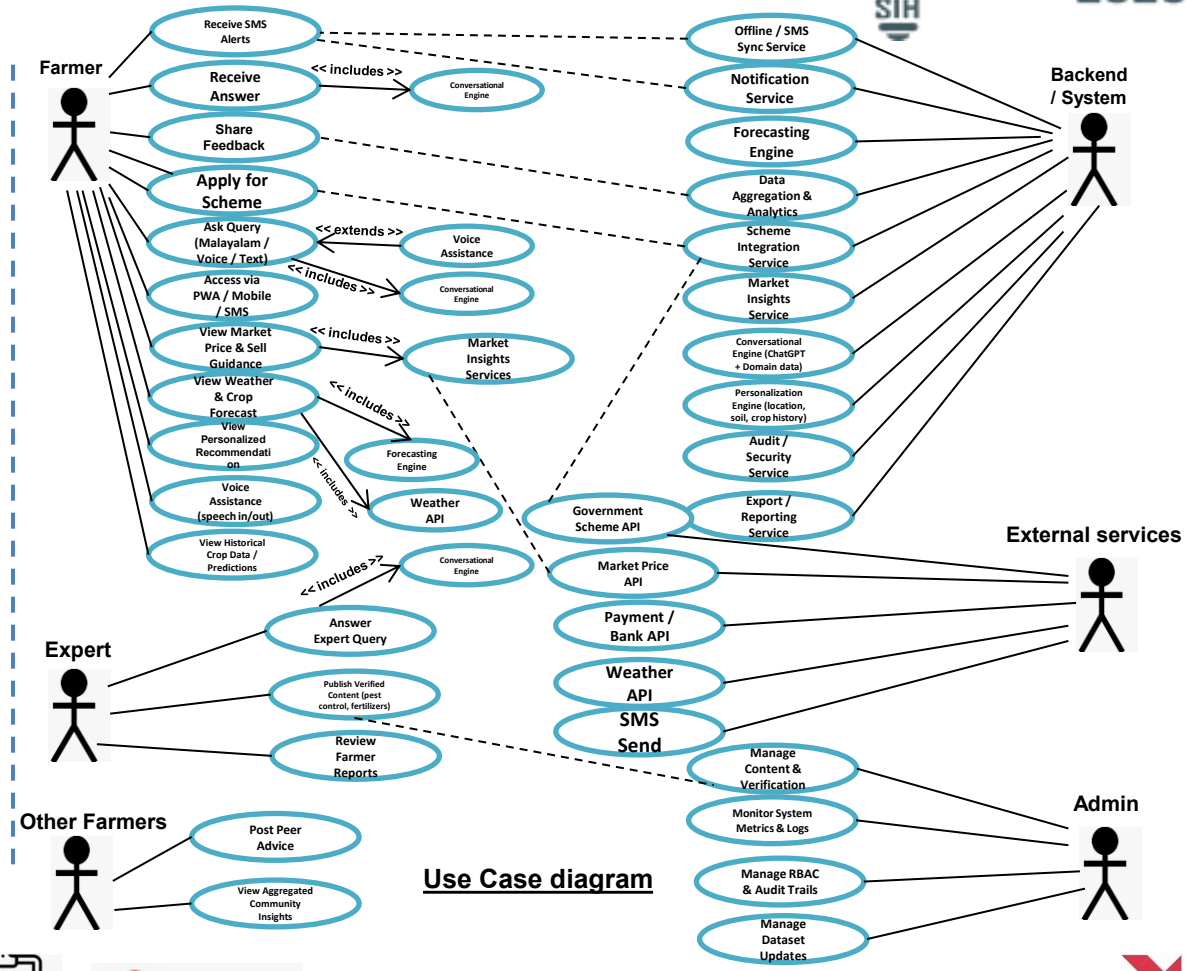
- AI-Powered Conversational Support
- Personalized Recommendations
- Inclusive Access Channels
- Data-Driven Market Insights

AI / ML services
- ChatGPT (GPT-5 mini),
- Python, scikit-learn,
- TensorFlow, Crop/Yield prediction,
- Weather forecasting

Backend
- Express, Node.js, Django,
- FastAPI, MongoDB, Firebase,
- GraphQL, JWT, RBAC,
- API versioning, Rate limiting

Front-End
- React.js, React Native,
- React Router, React query,
- HTML/CSS/JS (ES6+),
- Tailwind, Context API, REST,
- WebSocket, PWA

Offline & Edge
- Service Workers, IndexedDB
- SMS Alerts, Community Kiosks
- Offline Sync



Use Case diagram

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FEASIBILITY AND VIABILITY

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⚖️ Feasibility :

1. Combines schemes like Farm Plan, PKVY, PM-KISAN, PMKSY, overcoming delays and bureaucracy.
2. Leverages KSRTC buses, ferries, and trains for cheap, fast transport of produce and inputs.
3. Provides real-time alerts on crop failure, market prices, and insurance claims.
4. Strengthens decentralized governance by simplifying procedures.

⚙️ Viability :

1. Offers loans, subsidies, and direct income support to reduce debt and distress sales.
2. Connects farmers to insurance and irrigation schemes, managing climate risks.
3. Promotes organic farming, diversification, and precision irrigation for growth.
4. Supports social welfare to address farmer suicides and aging challenges.

⚠️ Challenges :

1. Complex procedures and overlapping orders delay scheme execution.
2. Digital literacy and connectivity gaps may hinder adoption.
3. Fragmented landholdings complicate logistics and profitability.
4. Weather unpredictability and market volatility remain risks.

Solutions :

1. Unified Farmer Platform:
   Simplify scheme access, insurance, transport, and alerts through a single mobile platform.
2. Smart Farming Support:
   Provide data-driven planning, irrigation tools, and training to reduce risks and boost income.

📊 Use Cases :

1. Streamlines market access using transport networks, reducing wastage.
2. One-stop platform for loans, subsidies, and insurance.
3. Provides climate and crop advisories for better planning.
4. Offers elder care support and welfare services.

📈 Business Potentials :

1. Data-driven services for banks, insurers, and policymakers.
2. Access to organic and premium markets.
3. Subscription-based advisory, logistics, and risk management.
4. Creates rural employment and strengthens local economies.

⭐ Supporting Facts for Feasibility and Viability ⭐

• Kerala's schemes like **PM-KISAN (₹6,000/year)**, **Agri Loans (subsidized interest)**, **PMKSY (irrigation support)**, **PKVY (organic farming)**, and **Agricultural Modernisation (subsidies)** provide financial backing, enabling project adoption.

• **KSRTC buses**, **trains**, and **ferries (₹6 fare)** ensure low-cost transport access for farmers statewide.

• With **agriculture growth falling from 4.91% to 0.87%**, **high debt**, **crop failures**, and **suicides in Wayanad**, urgent intervention is needed.
This proves the project's feasibility and viability, as it aligns with available resources and urgent farmer needs.

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IMPACT AND BENEFITS

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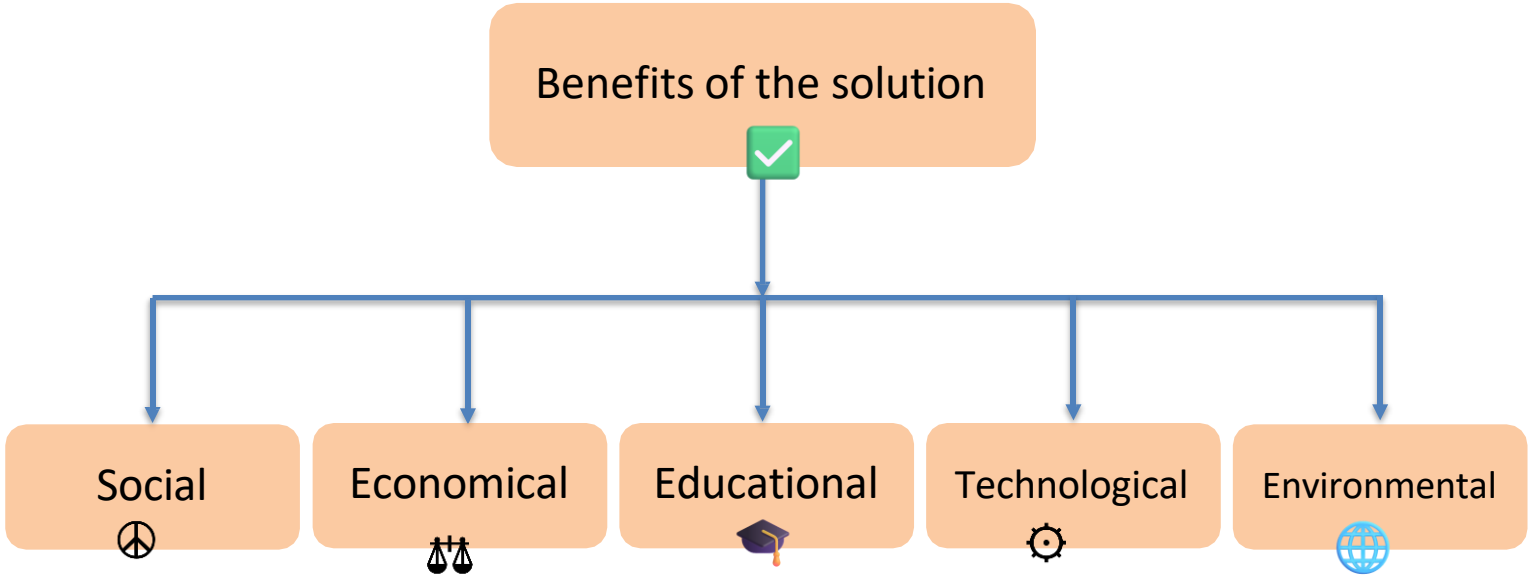

❖ Impact on Target Audience :

- Enables data-driven, climate-smart farming decisions.
- Builds trust through verified, government-backed advisory support.
- Enhances income with transparent pricing and crop planning tools.
- Expands access via voice, regional language, and keypad navigation.
- Promotes eco-friendly practices and efficient resource use.
- Strengthens climate resilience with timely alerts and guidance.
- Fosters community growth through expert networks and local logistics partnerships.

❖ Benefits of the solution :

Social Benefits :
- Builds trust with government-verified advisory services.
- Expands access via multilingual, voice tools for rural farmers.
- Encourages knowledge-sharing among farmers, transporters, and experts.

Economic Benefits :
- Stabilizes income with market-driven price forecasts.
- Ensures fair pricing by cutting middlemen.
- Creates rural jobs through transport empowerment.

Educational Benefits :
- Provides region-specific advice on soil, pests, and crops.
- Enhances skills and tech adoption for data-based farming.
- Bridges gaps with keypad and offline learning tools.

Technological Benefits :
- Uses AI for planning, pest control, and risk alerts.
- Offers real-time insights on weather and soil health.
- Delivers secure, scalable platforms that prevent fraud.

Environmental Benefits :
- Builds climate resilience with alerts for rainfall and droughts.
- Promotes sustainable farming and reduces soil loss.
- Optimizes water use and lowers environmental impact.

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RESEARCH AND REFERENCES

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❖ References – Types of Sources

Government Portals – Schemes, subsidies, stats

➤ keralaagriculture.gov.in, pmkisan.gov.in, pmksy.gov.in

Research Papers – Farmer distress, climate, crop failures

➤ scholar.google.com, researchgate.net

Market Data – Price trends, input costs, credit

➤ data.gov.in, https://agriwelfare.gov.in/

Weather & Disaster Info – Rainfall, droughts, floods

➤ mausam.imd.gov.in, https://sdma.kerala.gov.in/

AI/ML Tools – Forecasting, analytics

➤ https://www.tensorflow.org, https://scikit-learn.org/stable/, https://ai.google

NGOs & Reports – Field studies, farmer welfare

➤ fao.org, https://www.ifad.org/en/partners?utm_source=chatgpt.com,

Transport Data – Ferries, trains, costs

➤ https://ksrtcswift.kerala.gov.in/?utm_source=chatgpt.com, indianrail.gov.in, https://www.redbus.in/online-booking/ksrtc-kerala?utm_source=chatgpt.com,

Digital Inclusion – Internet access, literacy studies

➤ trai.gov.in, worldbank.org, undp.org

❖ Comparison with Existing Systems

Features	Our AI Advisory System	Existing Solutions
Multi-language support	✅ Malayalam & local dialects	❌ Limited languages
Real-time expert advice	✅ Instant AI-driven guidance	❌ Helplines delayed
Weather & crop alerts	✅ Time-series forecasts	❌ Manual alerts
Financial schemes access	✅ Integrated scheme info (PM-KISAN, PMKSY, loans)	❌ Scattered info
Market price trends	✅ Live updates from govt platforms	✅ Some access, irregular
Transport integration	✅ Suggests KSRTC, trains, ferries (₹6 fares)	❌ Not integrated
Insurance & subsidies	✅ Step-by-step guidance	✅ Limited access, complex
Social security	✅ Focus on aged farmers & credit	❌ Neglected
Community planning	✅ Verified data & local input	❌ Weak, bureaucratic
Digital access	✅ Offline-friendly, simple UI	❌ Internet-dependent

Video link : https://youtu.be/eHbQE1tBVcU?si=u5232Amb87LQuZ3L